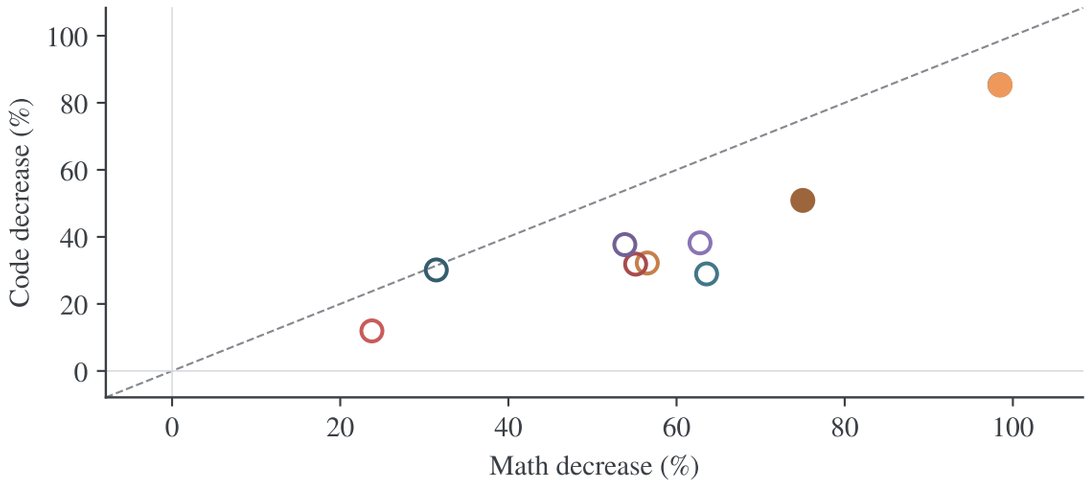
Llama-3.2-3B-Instruct | Held-out domain vs. domain

Relative RKL decrease (%)

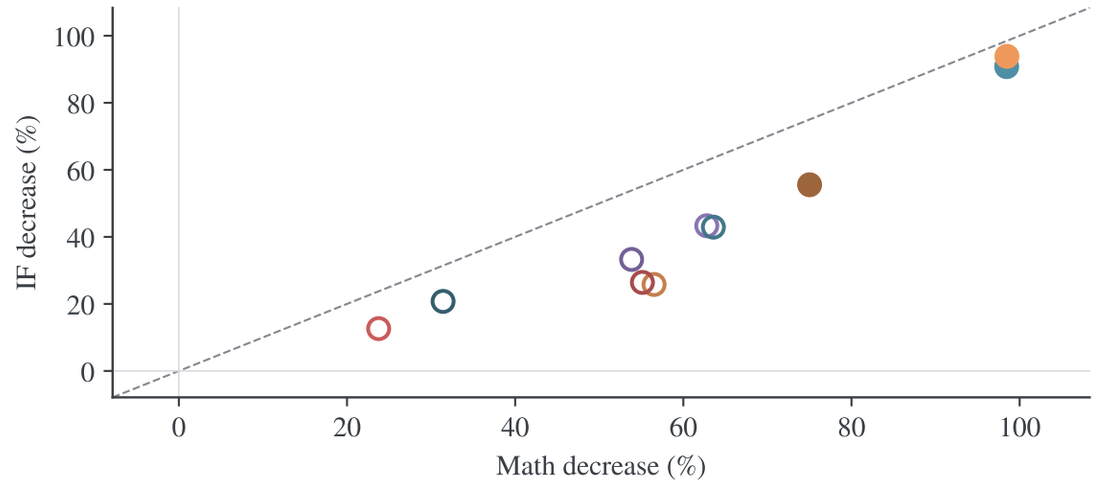
(a) Math vs. Code

n=10; r=0.93; rho=0.77



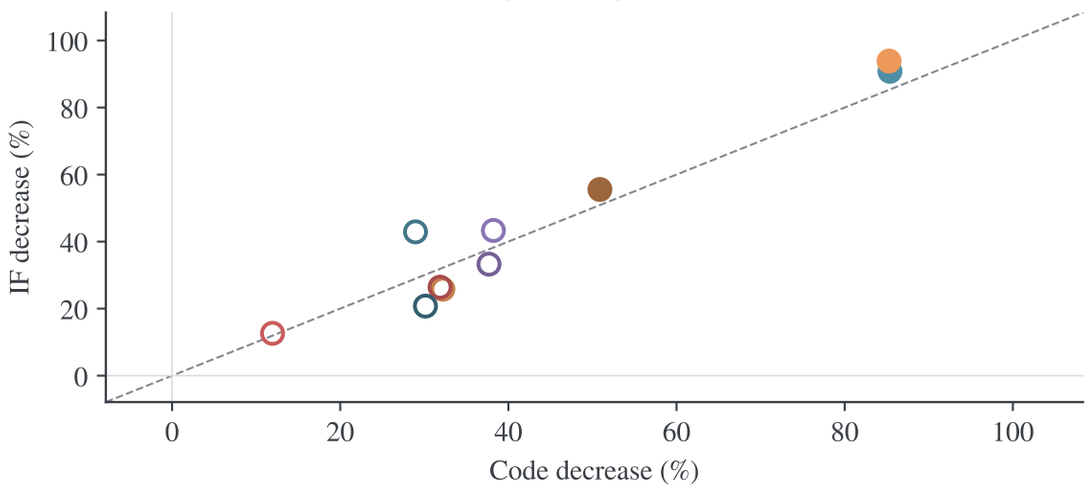
(b) Math vs. IF

n=10; r=0.95; rho=0.94



(c) Code vs. IF

n=10; r=0.97; rho=0.85



Co-rewarding-I-Llama-3.2-3B-Instruct-MATH [320]

GT-Llama-3.2-3B-Instruct-MATH [320]

Llama-3.1-8B-Instruct / Unsloth [320]

Llama-3.1-8B-SimpleRL-Zoo [320]

Llama-3.1-Nemotron-Nano-8B-v1 [320]

Llama-3.1-Tulu-3-8B [320]

Llama-3.1-Tulu-3-8B-DPO [320]

Llama-3.1-Tulu-3-8B-SFT [320]

Llama-3.2-1B-Instruct / Unsloth [320]

Llama-3.2-3B-Instruct-arithmetic-full [320]

Equal decrease

One point = one run endpoint; triangle: 64 updates; circle: 320; filled: same size; open: cross size.

Fixed-H RKL; dashed: equality; correlations pool all displayed endpoints (mixed budgets/sizes).